

SECURE CODING REVIEW REPORT

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Task:

Task 3 – Secure Coding Review

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1. Introduction

Secure coding is the practice of developing software applications in a way that protects them from security vulnerabilities and cyber-attacks. This report reviews a simple Python login application to identify security weaknesses and provide recommendations for improving its security. The objective is to understand common coding mistakes and demonstrate secure coding practices.

2. Objective

- Identify security vulnerabilities in the application.
- Analyse associated risks.
- Recommend secure coding practices.
- Provide an improved secure version of the application.

3. Application Under Review

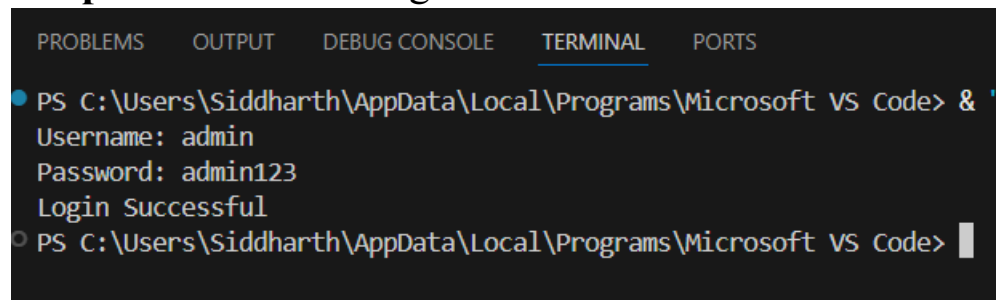
The application selected for review is a simple Python-based login system that verifies user credentials and grants access upon successful authentication.

Vulnerable Code:

```
username = input("Username: ")
password = input("Password: ")

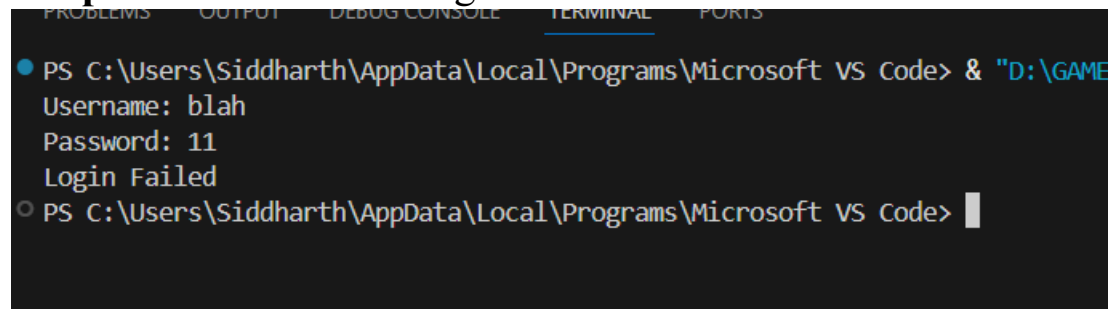
if username == "admin" and password == "admin123":
    print("Login Successful")
else:
    print("Login Failed")
```

Output: Successful Login

A screenshot of a terminal window in Visual Studio Code. The terminal shows the execution of a Python script. The prompt is 'PS C:\Users\Siddharth\AppData\Local\Programs\Microsoft VS Code> & ". The script prompts for 'Username: admin' and 'Password: admin123'. It then outputs 'Login Successful'. The prompt returns to 'PS C:\Users\Siddharth\AppData\Local\Programs\Microsoft VS Code>'.

```
PS C:\Users\Siddharth\AppData\Local\Programs\Microsoft VS Code> & "
Username: admin
Password: admin123
Login Successful
PS C:\Users\Siddharth\AppData\Local\Programs\Microsoft VS Code> |
```

Output: Unsuccessful Login

A screenshot of a terminal window in Visual Studio Code. The terminal shows the execution of a Python script. The prompt is 'PS C:\Users\Siddharth\AppData\Local\Programs\Microsoft VS Code> & "D:\GAME'. The script prompts for 'Username: blah' and 'Password: 11'. It then outputs 'Login Failed'. The prompt returns to 'PS C:\Users\Siddharth\AppData\Local\Programs\Microsoft VS Code>'.

```
PS C:\Users\Siddharth\AppData\Local\Programs\Microsoft VS Code> & "D:\GAME
Username: blah
Password: 11
Login Failed
PS C:\Users\Siddharth\AppData\Local\Programs\Microsoft VS Code> |
```

4. Code Analysis

The application was analysed manually to identify common security vulnerabilities and coding weaknesses.

VULNERABILITIES

5. Security Vulnerabilities Identified

5.1 Hardcoded Credentials

Description

The username and password are directly embedded within the source code.

Risk

Anyone with access to the source code can easily view the credentials.

Severity

High.

5.2 Plain Text Password Storage

Description

Passwords are stored and compared in plain text.

Risk

Attackers can obtain passwords if the code or storage location is exposed.

Severity

High.

5.3 Lack of Brute Force Protection

Description

The application allows unlimited login attempts.

Risk

Attackers can repeatedly guess passwords until successful.

Severity

Medium

Vulnerability Summary Table

Vulnerability	Severity
Hardcoded Credentials	High
Plain Text Password Storage	High
No Brute Force Protection	Medium

6. Security Recommendations

Recommendation 1

Use password hashing techniques such as SHA-256 or crypt.

Recommendation 2

Avoid storing credentials directly in source code.

Recommendation 3

Implement account lockout mechanisms after multiple failed login attempts.

Recommendation 4

Apply secure authentication practices.

Recommendation 5

Regularly perform security audits and code reviews.

SECURE CODE:

```
import hashlib

user = "admin"
stored_password = hashlib.sha256("admin123".encode()).hexdigest()

username = input("Username: ")
password = hashlib.sha256(input("Password: ").encode()).hexdigest()

if username == user and password == stored_password:
    print("Login Successful")
else:
    print("Login Failed")
```

OUTPUT: Successful Login

```
● PS C:\Users\Siddharth\AppData\Local\Programs\Microsoft VS Code> & "D:\GAMES\py
  Username: admin
  Password: admin123
  Login Successful
○ PS C:\Users\Siddharth\AppData\Local\Programs\Microsoft VS Code> █
```

7. Conclusion

The reviewed login application contained multiple security weaknesses, including hardcoded credentials, plain text password handling, and lack of brute force protection. Appropriate recommendations were provided and a more secure implementation was demonstrated. Following secure coding practices significantly reduces security risks and improves application resilience against cyber threats.